LIFTING ALGEBRA MAPS THROUGH POWERS OF A PRINCIPAL IDEAL WITH A NONZERODIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof identifies the coefficient shift through the nonzerodivisor property, pushes the obstruction past a^{n-c} using the annihilation hypothesis, observes the exponent grows, iterates, and takes the limit in the complete ring

Editorial note. The label lemma-get-morphism-nonzerodivisor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the obstruction class o(psi_n) in Ext^1_B(NL^wedge_{B/A}, a^nC/a^{2n}C) attached to a map psi_n : B -> C/a^nC, from the same chapter
- multiplication by a^c on a^nC/a^{2n}C identifies with the projection to a^{n-c}C/a^{2n-c}C when a is a nonzerodivisor on C
- the conversion of vanishing obstruction classes into lifted maps of Remark improve-homomorphism of the same chapter
- I-adic completeness of C and limits of compatible algebra maps

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 1244; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAP: sources/stacks-restricted/pinned-original-source.tex, source lines 1101–1101. Prerequisite pointer invoked at proof line(s) 1259; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $I = (a)$ be a principal ideal of a Noetherian ring A. Let B be an object of (0AL4). Assume given an integer $c \geq 0$ such that $\mathrm{Ext}^1_B(NL^\wedge_{B/A}, N)$ is annihilated by a^c for all B-modules N. Let C be an I-adically complete A-algebra such that a is a nonzerodivisor on C. Let $n > 2c$. For any A-algebra map $\psi_n : B \to C/a^nC$ there exists an A-algebra map $\varphi : B \to C$ such that $\psi_n \bmod a^{n-c}C = \varphi \bmod a^{n-c}C$.*

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Consider the obstruction class

$$o(\psi_n) \in \mathrm{Ext}^1_B(NL^\wedge_{B/A}, a^nC/a^{2n}C)$$

of Remark 0GAP. Since a is a nonzerodivisor on C the map $a^c : a^nC/a^{2n}C \to a^nC/a^{2n}C$ is isomorphic to the map $a^nC/a^{2n}C \to a^{n-c}C/a^{2n-c}C$ in the category of C-modules. Hence by our assumption on $NL^\wedge_{B/A}$ we conclude that the class $o(\psi_n)$ maps to zero in

$$\mathrm{Ext}^1_B(NL^\wedge_{B/A}, a^{n-c}C/a^{2n-c}C)$$

and a fortiori in

$$\mathrm{Ext}^1_B(NL^\wedge_{B/A}, a^{n-c}C/a^{2n-2c}C)$$

By the discussion in Remark 0GAP we obtain a map

$$\psi_{2n-2c} : B \to C/a^{2n-2c}C$$

which agrees with ψ_n modulo $a^{n-c}C$. Observe that $2n - 2c > n$ because $n > 2c$.

We may repeat this procedure. Starting with $n_0 = n$ and $\psi^0 = \psi_n$ we end up getting a strictly increasing sequence of integers

$$n_0 < n_1 < n_2 < \ldots$$

and A-algebra homorphisms $\psi^i : B \to C/a^{n_i}C$ such that ψ^{i+1} and ψ^i agree modulo $a^{n_i-c}C$. Since C is I-adically complete we can take φ to be the limit of the maps $\psi^i \bmod a^{n_i-c}C : B \to C/a^{n_i-c}C$ and the lemma follows. □